Security analysis of GroupTEE

1 Introduction

In this section, we present a formal security analysis of the proposed protocol in the Random Oracle Model (ROM). Authentication is proven under the Existential Unforgeability against Chosen-message Attacks (EUF-CMA) of the Schnorr signature scheme, while session-key confidentiality is established under standard Diffie-Hellman assumptions in the ROM via a real-or-random (ROR) indistinguishability experiment. The formal proofs are further complemented by symbolic analysis using BAN logic and automated verification with the Scyther tool. In addition, we provide an informal discussion covering replay resistance, forward and backward secrecy, and key agreement consistency.

2 Theoretical Proof

2.1 Authentication Proof

Assumptions. We work in a cyclic group $\mathbb{G}$ of prime order q generated by G. Each participant P_i holds a long-term secret key $sk_i \in \mathbb{Z}_q^*$ and the corresponding public key $pk_i = sk_i \cdot G$. The hash function $\mathsf{H} : \{0,1\}^* \to \mathbb{Z}_q$ is modeled as a random oracle. For the reduction, we assume access to a standard *single-signature Schnorr* signing oracle for a designated target identity, as required by the EUF-CMA security definition.

Honest Non-Colluding Model. We consider an honest-but-non-colluding execution model. All participating entities are assumed to follow the protocol specification faithfully and to generate authentication messages strictly according to the Schnorr signing procedure using fresh randomness and their own secret keys. No participant deviates from the prescribed message-generation algorithm, and no subset of participants colludes with the adversary to manipulate or forge authentication transcripts. Long-term secret keys remain confidential and are never revealed to other participants or to the adversary.

Theorem 1 (EUF-CMA authenticity of first-round messages).
Assume the hash function H *is modeled as a random oracle. If there exists a PPT adversary* $\mathcal{A}$ *that makes some honest participant accept a forged first-round message with non-negligible probability, then there exists a PPT reduction* $\mathcal{F}$ *that breaks the EUF-CMA security of the* single-signature *Schnorr scheme with non-negligible advantage.*

Proof. Define the signed message for m_i as $\mu_i := (ID_i \parallel R_{ix} \parallel R_{iy} \parallel \mathsf{Encode}(ID_{list}) \parallel sid)$, and the corresponding Schnorr signature as $\sigma_i^{(1)} = (R_i, s_i)$.

Game G0 Real execution. We consider the real protocol execution, where $\mathcal{A}$ may intercept, modify, or inject first-round messages $m_j = (ID_j, R_j, s_j)$.

Game G1 Embedding a Schnorr EUF-CMA challenge. $\mathcal{F}$ guesses a target identity index $j^\star \leftarrow \{1, \ldots, n\}$. For all $j \neq j^\star$, $\mathcal{F}$ generates (sk_j, pk_j) honestly and simulates the protocol. For the target $j^\star$, $\mathcal{F}$ sets $pk_{j^\star}$ to be the public key from the Schnorr EUF-CMA challenger and answers every protocol-generated signature of the form $\mathsf{Sign}(sk_{j^\star}, \mu)$ using the signing oracle. Hash queries are answered via a random-oracle table.

Game G2 Extracting a forgery. Suppose $\mathcal{A}$ succeeds in making some honest neighbor accept a first-round message $\hat{m}_{j^\star} = \{ID_{j^\star}, \hat{R}_{j^\star x}, \hat{R}_{j^\star y}, \hat{s}_{j^\star}\}$ that verifies under $pk_{j^\star}$ for the message $\hat{\mu}_{j^\star} = (ID_{j^\star} \parallel \hat{R}_{j^\star x} \parallel \hat{R}_{j^\star y} \parallel \mathsf{Encode}(ID_{list}) \parallel sid)$.

If $\hat{\mu}_{j^\star}$ was never queried to the signing oracle in Game 1, then $(\hat{\mu}_{j^\star}, (\hat{R}_{j^\star}, \hat{s}_{j^\star}))$ is a valid Schnorr EUF-CMA forgery and $\mathcal{F}$ outputs it.

Otherwise, $\hat{m}_{j^\star}$ is merely a replay of a previously issued protocol signature for the same (ID_{list}, sid) and does not constitute a new-message forgery. In that case, $\mathcal{F}$ aborts.

By the standard index-guessing argument,

$$\mathsf{Adv}_{\mathsf{1st\text{-}auth}}(\mathcal{A}) \;\leq\; n \cdot \mathsf{Adv}^{\mathsf{EUF\text{-}CMA}}_{\mathsf{Schnorr}}(\mathcal{F}) \;+\; \mathsf{negl}(\lambda).$$

Since Schnorr is EUF-CMA secure in the ROM [3], $\mathsf{Adv}_{\mathsf{1st\text{-}auth}}(\mathcal{A})$ is negligible.

Theorem 2 (EUF-CMA authenticity of second-round messages under batch verification).

Assume H *is modeled as a random oracle. If a PPT adversary* $\mathcal{A}$ *makes an honest participant* P_i *accept the second-round batch verification in Step 8 with non-negligible probability, then there exists a PPT reduction* $\mathcal{F}$ *that breaks the EUF-CMA security of the* single-signature *Schnorr scheme with non-negligible advantage, except with probability at most* $1/q$ *stemming from randomized batch verification.*

Proof. Define the signed message for M_j as $\nu_j := (ID_j \parallel X_{jx} \parallel X_{jy} \parallel R_{jx} \parallel R_{jy} \parallel \mathsf{Encode}(ID_{list}) \parallel sid)$, and the corresponding Schnorr signature as $\sigma_j^{(2)} = (R_j, t_j)$.

Game G0 Embedding a Schnorr EUF-CMA challenge. $\mathcal{A}$ interacts with honest parties and controls the network, aiming to make some honest P_i accept Step 8(4).

Game G1 Oracle setup and ROM programming. $\mathcal{F}$ guesses a target signer index $j^\star \leftarrow \{1, \ldots, n\}$. For all $j \neq j^\star$, $\mathcal{F}$ generates keys honestly and simulates the protocol. For $j^\star$, $\mathcal{F}$ sets $pk_{j^\star}$ to the challenger public key and answers every protocol-generated signature $\mathsf{Sign}(sk_{j^\star}, \nu)$ via the Schnorr signing oracle. Hash queries are answered via a ROM table.

Game G2 From acceptance to a Schnorr forgery. Suppose $\mathcal{A}$ makes some honest P_i accept Step 8(4), i.e., the randomized batch verification with fresh random weights $\{\alpha_j\}_{j \neq i}$ passes.

Let $\mathcal{E}$ be the event that *there exists* an index $j \neq i$ such that the *individual* Schnorr verification for $(\nu_j, (R_j, t_j))$ fails. By the standard soundness of randomized batch verification [1],

$$\Pr[\mathsf{Accept} \wedge \mathcal{E}] \leq 1/q.$$

Therefore, except with probability at most $1/q$, batch acceptance implies that all individual signatures are valid.

Condition on $\neg\mathcal{E}$. In particular, the individual verification for the target index $j^\star$ is valid, i.e., $(\hat{\nu}_{j^\star}, (\hat{R}_{j^\star}, \hat{t}_{j^\star}))$ verifies under $pk_{j^\star}$. If $\hat{\nu}_{j^\star}$ was never queried to the signing oracle, $\mathcal{F}$ outputs it as an EUF-CMA forgery. Otherwise it is a replay for the same message and $\mathcal{F}$ aborts.

Thus,

$$\mathsf{Adv}_{\mathsf{2nd\text{-}auth}}(\mathcal{A}) \;\leq\; n \cdot \mathsf{Adv}^{\mathsf{EUF\text{-}CMA}}_{\mathsf{Schnorr}}(\mathcal{F}) \;+\; \frac{1}{q} \;+\; \mathsf{negl}(\lambda).$$

Since Schnorr is EUF-CMA secure in the ROM, $\mathsf{Adv}_{\mathsf{2nd\text{-}auth}}(\mathcal{A})$ is negligible.

2.2 Confidentiality Proof

DDH Assumption. [2] Let $\mathbb{G}$ be a cyclic group of prime order q with generator G. Given a tuple $(G, A = aG, B = bG, C)$, where $a, b \in \mathbb{Z}_q^*$ are chosen randomly and C is either abG or a random element of $\mathbb{G}$, the Decisional Diffie-Hellman (DDH) assumption states that no polynomial-time adversary can distinguish between the two cases with non-negligible probability. Formally, for any PPT adversary $\mathcal{A}$:

$$\mathrm{Adv}^{\mathsf{DDH}}_{\mathcal{A}}(\lambda) = \Pr[\mathcal{A}(G, A, B, C) = 1] \leq \mathsf{negl}(\lambda),$$

where $\mathsf{negl}(\lambda)$ denotes a negligible function in the security parameter λ.

Theorem 3 (Session-key indistinguishability in the ROM under the DDH assumption).

Model H *and* KDF *as random oracles. Under the Decisional Diffie-Hellman (DDH) assumption in* $\mathbb{G}$*, the protocol achieves real-or-random (ROR) indistinguishability of the group session key* GK*.*

Proof. We consider a sequence of games $G0$-$G3$.

Game $G0$ *Real execution.* This is the real protocol and the standard ROR test: the adversary $\mathcal{A}$ receives either $GK = \mathsf{KDF}(\mathsf{EncodePoint}(K_i))$ or a uniformly random string of the same length. Let $p_0 = \Pr[b' = 1]$.

Game $G1$ *Simulating signatures.* We answer all Schnorr-related values (first-round and second-round messages) using the standard Schnorr simulation in the ROM, ensuring that all transmitted tuples verify correctly. This changes the view by at most a negligible amount, i.e., $|p_0 - p_1| \leq \mathsf{negl}(\lambda)$.

Game G2 Embedding a DDH instance. Let $(G, A = aG, B = bG, C)$ be a DDH challenge where C is either abG or a random element in $\mathbb{G}$. We embed A and B into the protocol by setting

$$R_i := A = aG, \qquad R_{i-1} := B = bG,$$

for a fixed index i, while simulating all other ephemeral public keys honestly.

Recall from Step 9 that the group element K_i is computed as

$$K_i = nr_i \cdot R_{i-1} + (n-1)X_i + (n-2)X_{i+1} + \cdots + X_{i-2}.$$

In particular, the first term

$$nr_i \cdot R_{i-1}$$

is an explicit Diffie–Hellman product between the secret scalar r_i and the public point R_{i-1}. Under the embedding above, this term becomes

$$nr_i \cdot R_{i-1} = n \cdot (aG \cdot b) = nabG.$$

We now replace this specific DH term by $n \cdot C$, obtaining a modified group element

$$\widetilde{K}_i = nC + (n-1)X_i + (n-2)X_{i+1} + \cdots + X_{i-2}.$$

If $C = abG$, then $\widetilde{K}_i$ is distributed identically to the real K_i. If C is uniformly random in $\mathbb{G}$, then nC is also uniformly random in $\mathbb{G}$, and hence $\widetilde{K}_i$ is statistically independent of the remaining linear terms.

Therefore, distinguishing between the real K_i and $\widetilde{K}_i$ would yield a distinguisher for the DDH problem. By the DDH assumption, we conclude that

$$|p_1 - p_2| \leq \mathsf{negl}(\lambda).$$

Game G3 Random key. In the random-C branch of $G2$, the encoded point $\mathsf{EncodePoint}(\widetilde{K}_i)$ is information-theoretically independent of $\mathcal{A}$'s view except with negligible probability. Since KDF is a random oracle, the output GK is uniformly random unless $\mathcal{A}$ queries KDF at the exact input $\mathsf{EncodePoint}(\widetilde{K}_i)$, which happens with probability at most negligible. Thus $p_3 = 1/2$ and $|p_2 - p_3| \leq \mathsf{negl}(\lambda)$.

Conclusion. By the triangle inequality, $|p_0 - 1/2| \leq \mathsf{negl}(\lambda)$.

2.3 key consistency

In the proposed protocol, each participant derives the final session key locally using broadcast parameters and its own private randomness, eliminating the need for centralized coordination or a trusted third party. Signature verification in both rounds guarantees that all parties maintain a consistent view of R_i and X_i, while any forged or inconsistent intermediate values are detected and discarded

during the verification phase. Using the following key computation formula, it is straightforward to prove that the protocol ensures correctness. Consequently, under honest execution, all honest parties compute the same session key, which guarantees both correctness and key consistency.

$$\begin{aligned}
K_i =& nr_i \cdot R_{i-1} + (n-1) \cdot X_i + (n-2) \cdot X_{i+1} + \cdots + X_{i-2} \\
=& r_i \cdot R_{i-1} + (r_i \cdot R_{i-1} + X_i) + (r_i \cdot R_{i-1} + X_i + X_{i+1}) + \cdots + \\
& (r_i \cdot R_{i-1} + X_i + \cdots + X_{i-2}) \\
=& r_i \cdot R_{i-1} + (r_i \cdot R_{i-1} + r_i \cdot R_{i+1} - r_i \cdot R_{i-1}) + \\
& (r_i \cdot R_{i-1} + r_i \cdot R_{i+1} - r_i \cdot R_{i-1} + r_{i+1} \cdot R_{i+2} - r_{i+1} \cdot R_i) + \cdots + \\
& (r_i \cdot R_{i-1} + r_i \cdot R_{i+1} - r_i \cdot R_{i-1} + \cdots + r_{i-2} \cdot R_{i-1} - r_{i-2} \cdot R_{i-3}) \\
=& r_i \cdot R_{i-1} + r_i \cdot R_{i+1} + r_{i+1} \cdot R_{i+2} + \cdots + r_{i-2} \cdot R_{i-1} \\
=& r_i \cdot R_{i+1} + r_{i+1} \cdot R_{i+2} + \cdots + r_{i-2} \cdot R_{i-1} + r_{i-1} \cdot R_i
\end{aligned}$$

where $X_i = r_i \cdot (R_{i+1} - R_{i-1})$. The right-hand side is a cyclically invariant symmetric sum, and therefore it follows that $K_i = K_j$ for any i, j. By the deterministic nature of the key derivation function, it follows that $GK_i = GK_j$, that is, all honest parties derive the same group session key.

3 Formal Security Analysis

3.1 BAN Logic Analysis

BAN logic is employed to analyze the authentication and key-agreement beliefs established by GroupTEE. The purpose of this analysis is not to verify algebraic correctness of the key computation, but to reason about belief establishment and mutual authentication.

The analysis proceeds in four stages: (1) formalization of security goals, (2) idealization of protocol messages, (3) definition of initial assumptions, (4) derivation using BAN inference rules.

We emphasize that BAN logic abstracts Schnorr signatures as ideal authentication primitives. Since all signed messages in GroupTEE include the session identifier *sid*, freshness is bound to the session context rather than to timestamps.

The following BAN inference rules are used.

Jurisdiction Rule:

$$\frac{P \mid\equiv Q \Rightarrow X \quad P \mid\equiv Q \mid\equiv X}{P \mid\equiv X} \tag{1}$$

Message-Meaning Rule (Signature Form):

$$\frac{P \mid\equiv \mathrm{pub}_Q \leftrightarrow Q \quad P \lhd \{X\}_{sk_Q}}{P \mid\equiv Q \mid\sim X} \tag{2}$$

Nonce-Verification Rule:

$$\frac{P \mid\equiv \#(X) \quad P \mid\equiv Q \mid\sim X}{P \mid\equiv Q \mid\equiv X} \tag{3}$$

Formal Proof Goal The goal of GroupTEE is for all legitimate participants $P_1, \ldots, P_n$ to achieve mutual authentication and to agree on the same fresh group session key after two rounds of interaction. Since GroupTEE derives a single group key GK shared by all members in the authenticated participant list, we specify the following BAN-logic goals for any two distinct members P_i and P_j $(i \neq j)$:

Goal1 (Key belief). $P_i \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

Goal2 (Key belief). $P_j \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

Goal3 (Mutual belief). $P_i \mid\equiv P_j \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

Goal4 (Mutual belief). $P_j \mid\equiv P_i \mid\equiv \big(P_i \overset{GK}{\longleftrightarrow} P_j\big)$.

In addition, we require freshness of the established group key to be bound to the session context:

Goal5 (Key freshness). $P_i \mid\equiv \#(GK)$ and $P_j \mid\equiv \#(GK)$.

Idealized Protocol Let $\{X\}_{sk_i}$ denote a Schnorr signature generated using P_i's private key. The session identifier sid is included in all signed messages and serves as the freshness indicator of the protocol session.

The idealized messages are:

Message 1 (First Round).

$$P_i \rightarrow P_{i\pm1} : P_{i\pm1} \lhd \{ID_i, R_i, sid\}_{sk_i}$$

Message 2 (Second Round Broadcast).

$$P_i \rightarrow P_j \ (\forall j \neq i) : P_j \lhd \{ID_i, X_i, R_i, sid\}_{sk_i}$$

Initial Assumptions Before protocol execution, we formalize the trust relationships, freshness conditions, and key-derivation properties that are assumed to hold among honest participants.

A1: $P_i \mid\equiv CA \Rightarrow (\mathrm{pub}_k \leftrightarrow ID_k)\ (\forall k)$
A2: $P_i \mid\equiv CA \mid\equiv (\mathrm{pub}_k \leftrightarrow ID_k)\ (\forall k)$
A3: $P_i \mid\equiv \#(sid)$
A3': $P_i \mid\equiv \#(sid) \Rightarrow \#(ID_k, R_k, sid) \ \wedge\ \#(ID_k, X_k, R_k, sid)\ (\forall k)$
A4: $P_i \mid\equiv \#(R_k)$
A5: $P_i \mid\equiv \#(X_k)$
A6: $P_i \mid\equiv GK = \mathsf{KDF}(f(R_1, \ldots, R_n, X_1, \ldots, X_n, sid))$

Derivation Consider two arbitrary participants P_i and P_j $(i \neq j)$.

Step 1: Public-key binding.

From A1 and A2 and applying the Jurisdiction rule, participant P_j believes:

$$P_j \mid\equiv (\mathrm{pub}_i \leftrightarrow ID_i).$$

Step 2: First-round authentication.

Upon receiving Message 1 and applying the Message-Meaning rule:

$$P_j \mid\equiv P_i \mid\sim (ID_i, R_i, sid).$$

Since sid is fresh (A3), by Nonce Verification:

$$P_j \mid\equiv P_i \mid\equiv (ID_i, R_i).$$

Thus P_j believes R_i originates from P_i in this session.

Step 3: Second-round authentication.

Upon receiving Message 2 and applying Message-Meaning:

$$P_j \mid\equiv P_i \mid\sim (ID_i, X_i, R_i, sid).$$

Using freshness of sid again:

$$P_j \mid\equiv P_i \mid\equiv (ID_i, X_i, R_i).$$

Therefore, P_j believes that X_i and R_i are fresh and generated by P_i.

Step 4: Key agreement belief.

Since all participants derive the session key from the same authenticated set $\{(ID_k, R_k, X_k, sid)\}_{k=1}^{n}$ and believe in the correctness of the key-derivation function (A6), it follows that:

$$P_i \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j,$$

$$P_j \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j.$$

Furthermore, since each party believes that the other party believes in the same authenticated parameters, we obtain:

$$P_i \mid\equiv P_j \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j,$$

$$P_j \mid\equiv P_i \mid\equiv P_i \overset{GK}{\longleftrightarrow} P_j.$$

Therefore, under the stated assumptions, GroupTEE achieves mutual authentication and shared session-key belief among all honest participants. Each party believes that the established key is fresh, authenticated, and shared exclusively with legitimate group members.

3.2 Formal Analysis Tool

Scyther is employed to perform symbolic verification of GroupTEE under the Dolev–Yao adversary model.In the Scyther specification, Schnorr signatures are abstracted as ideal digital signatures $\{m\}_{\mathrm{sk}}$ that are publicly verifiable via the corresponding public key, captured by the relation inversekeys(pk, sk). The session identifier sid, which (abstracts the timestamp/session context in the protocol, is included in the signed payload to enforce session binding and to prevent replay across sessions.

We specify standard authentication claims Alive, Weakagree, Nisagree, Nisyncn for each role and a secrecy claim on an abstract session key derived from the authenticated broadcast parameters of both rounds. Fig. 1 illustrates the Scyther-based formal modeling of the GroupTEE protocol, and the results show that all specified security claims are successfully satisfied within the given bounded analysis scope, as illustrated in Fig. 2.The authentication- and synchronization-related claims, including Alive, Weakagree, Niagree, and Nisynch, are all marked as Ok, indicating that the protocol is resilient to replay, impersonation, and desynchronization attacks within the current model and search bounds.

References

1. Bellare, M., Garay, J.A., Rabin, T.: Fast batch verification for modular exponentiation and digital signatures. In: International conference on the theory and applications of cryptographic techniques. pp. 236–250. Springer (1998)
2. Boneh, D., Shoup, V.: A graduate course in applied cryptography. Draft 0.5 p. 14 (2020)
3. Pointcheval, D., Stern, J.: Security proofs for signature schemes. In: International conference on the theory and applications of cryptographic techniques. pp. 387–398. Springer (1996)

```
hashfunction H;
const pkA, pkB, pkC: Function;
secret skA, skB, skC: Function;
inversekeys (pkA,skA);
inversekeys (pkB,skB);
inversekeys (pkC,skC);

protocol SEAGKAP(A, B, C)
{
  role A
  {
    fresh rA: Nonce;
    fresh RA: Nonce;
    var RB: Nonce;
    var RC: Nonce;
    fresh sid: Nonce;
    send_1(A, B, A, RA, sid, {A,RA,sid}skA);
    send_2(A, C, A, RA, sid, {A,RA,sid}skA);
    recv_3(B, A, B, RB, sid, {B,RB,sid}skB);
    recv_5(C, A, C, RC, sid, {C,RC,sid}skC);

    fresh XA: Nonce;
    var XB: Nonce;
    var XC: Nonce;
    send_7(A, B, A, XA, sid, {A,XA,sid}skA);
    send_8(A, C, A, XA, sid, {A,XA,sid}skA);
    recv_9(B, A, B, XB, sid, {B,XB,sid}skB);
    recv_11(C, A, C, XC, sid, {C,XC,sid}skC);

    claim(A, Secret, H(rA, RC, XA, XB));
    claim(A, Alive);
    claim(A, Weakagree);
    claim(A, Niagree);
    claim(A, Nisynch);
  }

  role B
  {
    fresh RB: Nonce;
    fresh rB: Nonce;
    var RA: Nonce;
    var RC: Nonce;
    var sid: Nonce;
    recv_1(A, B, A, RA, sid, {A,RA,sid}skA);
    send_3(B, A, B, RB, sid, {B,RB,sid}skB);
    send_4(B, C, B, RB, sid, {B,RB,sid}skB);
    recv_6(C, B, C, RC, sid, {C,RC,sid}skC);

    fresh XB: Nonce;
    var XA: Nonce;
    var XC: Nonce;
    recv_7(A, B, A, XA, sid, {A,XA,sid}skA);
    send_9(B, A, B, XB, sid, {B,XB,sid}skB);
    send_10(B, C, B, XB, sid, {B,XB,sid}skB);
    recv_12(C, B, C, XC, sid, {C,XC,sid}skC);

    claim(B, Secret, H(rB, RA, XB, XC));
    claim(B, Alive);
    claim(B, Weakagree);
    claim(B, Niagree);
    claim(B, Nisynch);
  }
  role C
  {
    fresh rC: Nonce;
    fresh RC: Nonce;
    var RA: Nonce;
    var RB: Nonce;
    var sid: Nonce;
    recv_2(A, C, A, RA, sid, {A,RA,sid}skA);
    recv_4(B, C, B, RB, sid, {B,RB,sid}skB);
    send_5(C, A, C, RC, sid, {C,RC,sid}skC);
    send_6(C, B, C, RC, sid, {C,RC,sid}skC);

    fresh XC: Nonce;
    var XA: Nonce;
    var XB: Nonce;
    recv_8(A, C, A, XA, sid, {A,XA,sid}skA);
    recv_10(B, C, B, XB, sid, {B,XB,sid}skB);
    send_11(C, A, C, XC, sid, {C,XC,sid}skC);
    send_12(C, B, C, XC, sid, {C,XC,sid}skC);
    claim(C, Secret, H(rC, RB, XC, XA));
    claim(C, Alive);
    claim(C, Weakagree);
    claim(C, Niagree);
    claim(C, Nisynch);
  }
}
```

Fig. 1. Scyther-based formal model of GroupTEE

Scyther results : verify

Claim				Status	Comments
SEAGKAP	A	SEAGKAP,A1	Secret H(rA,RC,XA,XB)	Ok	No attacks within bounds.
		SEAGKAP,A2	Alive	Ok	No attacks within bounds.
		SEAGKAP,A3	Weakagree	Ok	No attacks within bounds.
		SEAGKAP,A4	Niagree	Ok	No attacks within bounds.
		SEAGKAP,A5	Nisynch	Ok	No attacks within bounds.
	B	SEAGKAP,B1	Secret H(rB,RA,XB,XC)	Ok	No attacks within bounds.
		SEAGKAP,B2	Alive	Ok	No attacks within bounds.
		SEAGKAP,B3	Weakagree	Ok	No attacks within bounds.
		SEAGKAP,B4	Niagree	Ok	No attacks within bounds.
		SEAGKAP,B5	Nisynch	Ok	No attacks within bounds.
	C	SEAGKAP,C1	Secret H(rC,RB,XC,XA)	Ok	No attacks within bounds.
		SEAGKAP,C2	Alive	Ok	No attacks within bounds.
		SEAGKAP,C3	Weakagree	Ok	No attacks within bounds.
		SEAGKAP,C4	Niagree	Ok	No attacks within bounds.
		SEAGKAP,C5	Nisynch	Ok	No attacks within bounds.

Done.

Fig. 2. Scyther-based formal analysis result of GroupTEE